Dylan Matwe

Engineering Portfolio

B.Sc. Mechanical Engineering, Minor in Aerospace Engineering | University of Calgary
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Highlights

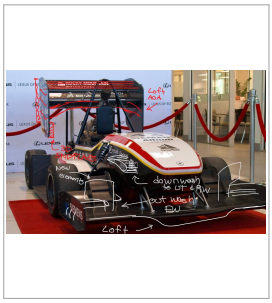
+45%	**1st of 70**	**−75%**	**−60%**
full-car downforce (+15% drag)	teams in aerodynamics design, FSAE Electric 2024	CFD meshing time, Hitachi Energy plugin	aero element composite build time

UCalgary Racing – Formula SAE Electric

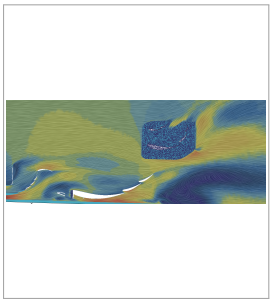
Aerodynamics Team Lead — Sep 2023 – Oct 2024
Aerodynamics Sub-team — *Calgary, Alberta*

Full-Car Aerodynamics Package

UCR-01 | 2023/2024 Season



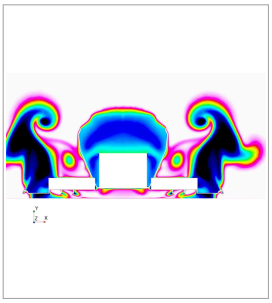
Concept



CFD (STAR-CCM+)



Surface pressure



Y250 vortex



Final car

Goals

- Raise full-car downforce by more than 40% over the SR-22 package with drag growth under 20%
- Develop the front wing, rear wing, and bodywork as one integrated package rather than as isolated components using full car simulations, enabled by HPC
- Use the front wing & bodywork to condition the downstream flow, manage the front tire wake and feed the undertray and rear wing
- Keep every component FSAE rules compliant, and buildable in-house to CAD tolerance
- Build sub-team capability in CFD-driven design through structured training and design reviews

Process

- **Front wing:** Designed the wing to condition the flow for the rest of the car, not only to generate its own downforce. It sheds a strong Y250 vortex that provides outwash, displacing the front tire wake outboard and away from the downstream aerodynamic surfaces.
- **Bodywork:** Led packaging and integration with the other sub-teams. Added whisker elements that downwash flow toward the undertray and rear wing, improving both. The whiskers create local lift but a net downforce gain of on each of the front and rear wings, at very low drag (efficiency of 4).
- **Rear wing:** Added two upper (bench) elements that ingest clean, undisturbed air for **+15%** rear wing downforce. Lofted an element so the sections near the endplates run a higher angle of attack in cleaner flow, delaying stall without leading-edge separation. Fitted strakes that extend the wing to the limits of the rules-defined bounding box, trimmed to exact dimensions at assembly to ensure compliance at technical inspection.
- **Methods:** Evaluated every component within the full-car STAR-CCM+ model to capture interaction effects, and carried each design from concept sketch through CFD to manufacture.
- **Leadership:** Trained 14 sub-team members on scoped components with guided CFD instruction, and instituted design reviews requiring aerodynamic justification and manufacturing trade-offs for every element. Lead to continued success in future seasons with consistent top 3 finishes in FSAE aerodynamics design by future teams since the 2023 season.

Outcome

- **+45%** full-car downforce for **+15%** drag
- **+50%** rear wing downforce
- **1st of 70 teams** in aerodynamics design, FSAE Electric 2024

Skills: STAR-CCM+, SOLIDWORKS, DFMA, Multi-element wing design, Vortex management, Technical leadership

UCalgary Racing – Formula SAE Electric

Aerodynamics Team Lead Sep 2023 – Oct 2024

Aerodynamics Sub-team *Calgary, Alberta*

Composite Aero Manufacturing

UCR-01 | 2023/2024 Season



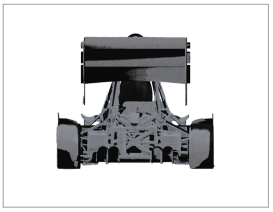
Mold



Wing in progress



Front & rear wings



3D scan vs. CAD



Completed car

Goals

- Cut build time per aero element by more than 50%
- Bring part geometry within a measurably tighter tolerance of CAD (10% or better)
- Make multi-element composite parts repeatable in-house

Process

- Integrated 3D-printed tooling and CNC-machined wood molds into the layup process for repeatable part geometry
- Expanded 3D-printed parts across the car where they shortened iteration time
- Verified each build against CAD with 3D scan comparison
- Gated designs on manufacturability: deferred those that could not be built well, and approved complex builds only where the performance gain justified them

Outcome

- **−60%** build time per aero element
- **+14%** dimensional accuracy

Skills: Composites, Tolerancing, CNC mold tooling, 3D printing, 3D scanning, DFM

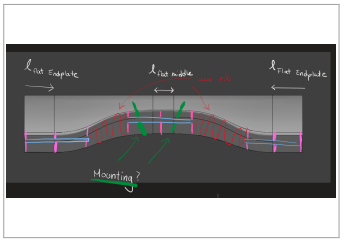
UCalgary Racing – Formula SAE Electric

Aerodynamics Team Member Oct 2024 – Present

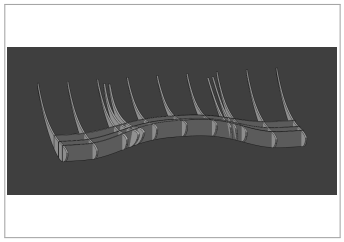
Aerodynamics Sub-team *Calgary, Alberta*

Rear Wing Package Upgrade

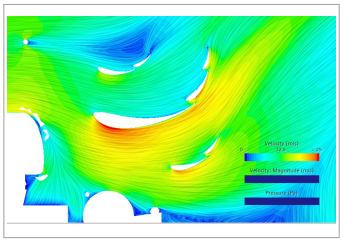
UCR-04 | 2026/2027 Season



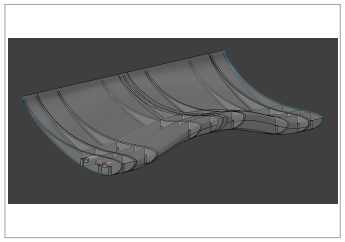
Concept sketch



Initial CAD



CFD



Final CAD

Goals

- Deliver a package-level rear wing upgrade without a full redesign
- Concentrate gains in element 1, the largest and highest-leverage element
- Increase manufacutring quality, reduce cost, and trial new aerostructural methods.

Process

- Targeted element 1 as the largest element and the greatest source of potential gain
- Replaced the purchased carbon tubes with in-house carbon and foam spars, which are lighter, accept any geometry, and reduce reliance on outside suppliers and shipping. This enabled a larger loft than the tubes had allowed
- Added a beam wing with extra elements that had been rejected for UCR-01 as too labour-intensive, but were now able to be remade with existing tooling. A deliberate trade tween manufacturing effort and performance

Outcome

- **−30%** cost
- **−15%** mass
- **+10%** downforce
- No reliance on external suppliers for carbon tubes
- Able to loft elements to significantly higher relative angles

Skills: SOLIDWORKS (parametric modeling), STAR-CCM+, DFMA, Lightweight structures

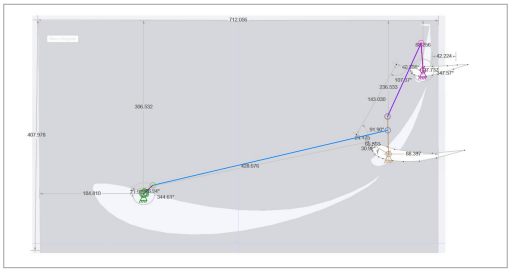
UCalgary Racing – Formula SAE Electric

Aerodynamics Team Member Sep 2022 – Sep 2023

Aerodynamics Sub-team *Calgary, Alberta*

Drag Reduction System Concept

SR-22 | 2022/2023 Season



DRS concept sketch

Goals

- Cut rear wing drag at high speed
- Deliver the team's first DRS concept, with an actuator and linkage compatible with the rear wing structure

Process

- Developed the concept in my first year on the team, while building CFD and SOLIDWORKS skills
- Selected the actuator, designed the linkage, and used CFD to evaluate the drag benefit of the open configuration

Outcome

- CFD predicts a **40%** reduction in rear wing drag at high speed
- Handed off at concept stage. Another team member later took it forward and built it for UCR-03 (2024/2025 Season)

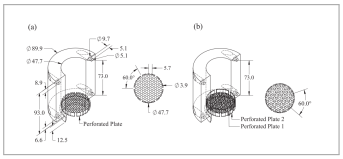
Skills: CFD, Mechanism and linkage design, Actuator selection, SOLIDWORKS

University of Calgary

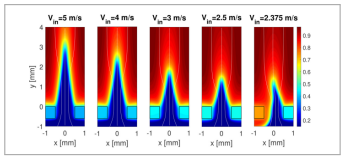
Summer Research Assistant May 2025 – Aug 2025

Reacting Flow and Numerical Methods Group *Calgary, Alberta*

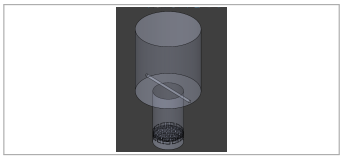
CFD Validation for Hydrogen Flashback



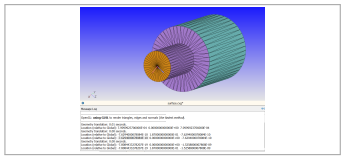
Experimental rig CAD



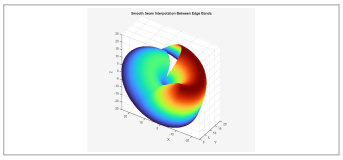
Experimental combustion



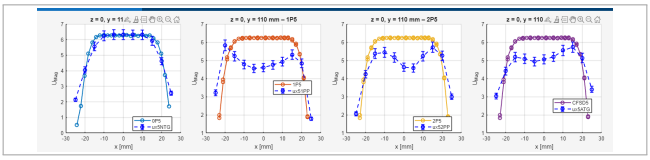
CFD domain



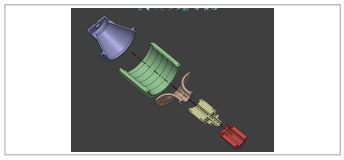
Setting up the CFD case



Generated inputs



Simulation vs. experiment across turbulence conditions



Full parametric geometry

Reference images from S. Mohammadnejad, L. Saca, and S. Kheirkhah, "A new compact active turbulence generator for premixed combustion: Non-reacting flow characteristics," *Physics of Fluids* 34, 105102 (2022), doi:10.1063/5.0108020.

Goals

- Reproduce the published non-reacting experiment velocity and pressure profiles
- Fix the turbulence settings and boundary conditions before moving to reacting flow
- Establish a validated baseline for the hydrogen flashback stage

Process

- Rebuilt the published turbulence-generator experiment in CONVERGE CFD, sweeping non-reacting cold-flow cases to calibrate turbulence and boundary conditions against experimental data
- Scripted the generation of simulation inputs, which made large parameter sweeps practical
- Rebuilt static object-file geometry as parametric SOLIDWORKS models and drawings, reducing solver and meshing crashes, as well as iteration time
- Purely computational: validated against published laboratory data, with no physical testing

Outcome

- Velocity and pressure profiles within **10%** error
- **−60%** iteration time and far fewer crashes

Skills: CONVERGE CFD, MATLAB, SOLIDWORKS, Turbulence modeling, Validation

Hitachi Energy

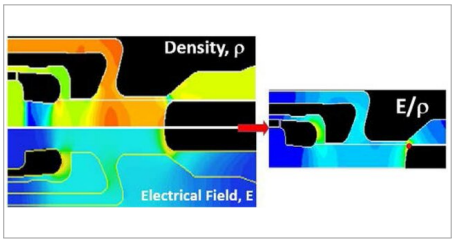
Simulations Intern Sep 2025 – Aug 2026

Circuit Breaker R&D *Baden, Switzerland*

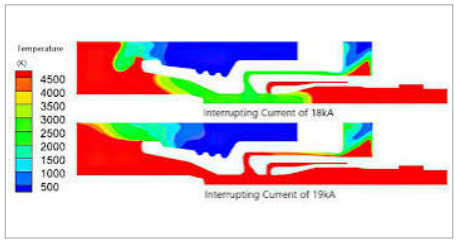
Transient Arc and Gas Flow CFD

245 kV EconiQ SF_6-free breaker

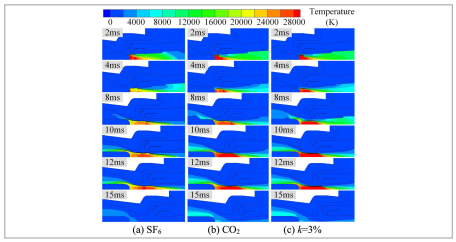
Work at Hitachi Energy is confidential. The CFD images below are representative examples from public sources, not my own simulations or data.



Representative: gas flow CFD



Representative: flow field



Representative: transient timesteps

Image source: *Arc Simulations of a 252 kV Self-blast Circuit Breaker for Different Currents Considering Gas Control Valves*, conference paper, Springer, 2022, pp. 405–413, doi:10.1007/978-981-19-1528-4_41, Fig. 5.

Goals

- Quantify how R&D design changes affect arc interruption performance
- Extend my non-reacting, low-speed flow experience (Formula SAE) and the reacting-flow work from my research to coupled simulation, with electrodynamics in place of combustion

Process

- Ran transient gas flow and arc simulations on HPC clusters, solving a reduced form of Maxwell's equations coupled to the flow
- Assessed dielectric recovery: the transient recovery voltage across the open contacts, residual charge in the gas, the likelihood of arc reignition as the gas cools, and the capacitance of the full assembly
- Disassembled, scanned, and documented lab-tested components, and compared ablation, temperature build-up, and pressure build-up against simulation

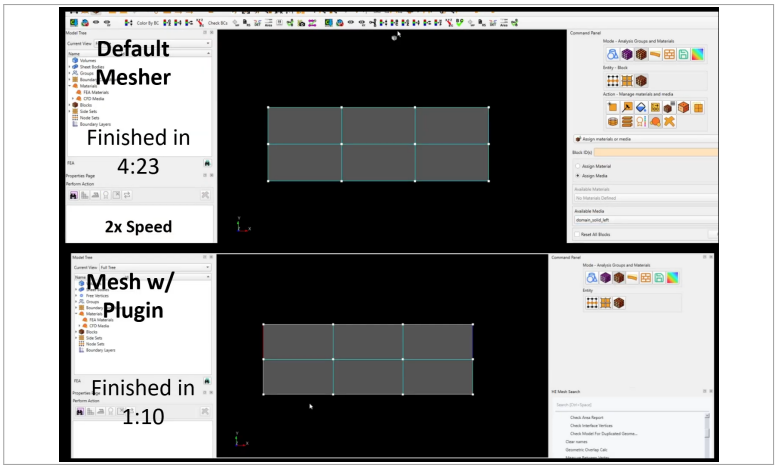
Outcome

- Design changes quantified for the 245 kV SF_6-free breaker allowing for on time project completion
- Validated simulations to within 10% with lab tested components
- Hands-on experience coupling flow and electrodynamics

Skills: Transient CFD, HPC, Multiphysics (flow and electrodynamics), Test correlation

Python Meshing Plugin for Coreform Cubit

Gas Circuit Breaker Team



Plugin in use

Goals

- Cut CFD meshing time by more by scripting the repetitive geometry creation, meshing, and naming workflow

Process

- Identified a gap in the team's meshing workflow and took the initiative to close it
- Studied the Cubit documentation and consulted the developers directly
- Designed and built the tool alone as a structured, extensible Python repository, developing program design and Python skills along the way

Outcome

- **−75%** CFD meshing time, measured on a reduced, non-confidential sample mesh. The plugin significantly speeds up the geometry, meshing, and naming processes, which are major steps of the meshing process at Hitachi
- Deployed to the 20-person circuit breaker team and all Cubit users company-wide
- An extensible base for new scripts, built independently from the first line

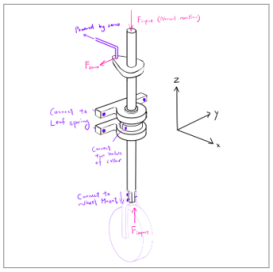
Skills: Python, Coreform Cubit, Software design, Automation, Independent initiative

ENME 493 Design Project 2024/2025

University of Calgary

Lightweight Leaf-Spring UAV Landing Gear

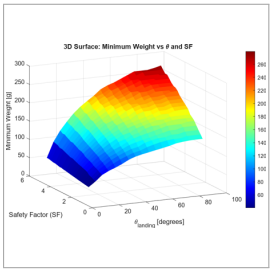
2025 SUAV Competition



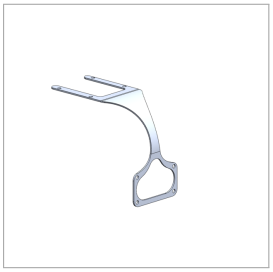
Inital steering sketch



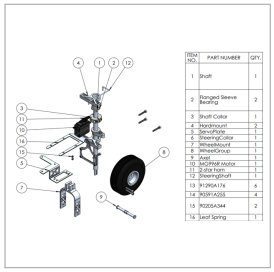
Original off-the-shelf spring



Min. weight vs. landing angle and SF



Final spring design



Assembly view

Goals

- Cut landing gear weight from the 521 g off-the-shelf unit, and cost from $106
- Right-size the structure: the previous gear was over-engineered at a factor of safety above 20
- Soften landings to protect avionics, and replace the unreliable servo steering
- Increase propeller clearance and fit the fuselage plate, with a design that suits both runway and rough-field landings

Process

- **Diagnosis:** Identified why the previous gear failed the team: stiffness that damaged avionics on landing, poor steering, low ground clearance, and poor fuselage fitment from using an off-the-shelf part.
- **Spring optimization:** Set up a multi-variable MATLAB optimization that minimizes spring weight and landing deceleration over geometry (fillet radius, thickness, width, base dimension) and material properties (E, S_u, ρ), subject to a Goodman fatigue safety factor above 2. Swept aluminum, steel, titanium, magnesium, and ABS candidates across landing angle.
- **Steering and tire assembly:** Designed a servo-driven steering shaft with flanged sleeve bearings, and analyzed worst-case loads (servo at maximum torque, bearing binding, highest impulse). Shaft factor of safety 19.7 (7075) and 10.8 (6061); axle 1.7 to 1.9 at yield under worst-case loading.
- **Manufacturing:** Designed every part for in-house production: rolled, punched, and bent sheet for the spring and servo plate, and a milled, drilled, and reamed shaft, reducing cost.

Outcome

- **−51%** weight (521 to 266.4 g)
- **−49%** cost ($106 to $52)
- Spring: **−87%** weight and **+65%** damping over the initial design
- Added adjustable propeller clearance, reliable servo steering, and a fuselage-plate mount; approved for manufacture for the 2025 SUAV competition

Skills: MATLAB (optimization), SOLIDWORKS, Fatigue and strength analysis, DFM, Leaf spring design